52 Archaeo.pro: an open, offline-first toolchain for the sorveglianza archeologica, with the archive owned by the archaeologist

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Preventive archaeology in Italy runs on paperwork. A sorveglianza archeologica means recording on site, often with no connectivity; gathering authoritative constraints from national and regional services; and delivering a report in the form the Soprintendenza expects. The data is produced once, delivered once, and then effectively lost, to the professional who made it as much as to research.

Archaeo.pro is an open-source toolchain built for that workflow rather than around it. An offline-first Angular PWA (MapLibre, local scratch storage) records contexts, geometries and photographs in the field; the same build ships as a Tauri 2 native application on iOS and Android. A FastAPI backend proxies the national geoportal's (PCN) WMS layers (orthophotos, IGM, geology, land cover) and renders a Soprintendenza-ready DOCX, converted to PDF by a self-hosted Gotenberg service.

The design decision we want to put to the panel is storage. Each archaeologist's data lives in their own GitHub account: a private index repository plus one repository per surveillance, photographs as release assets — under OAuth PKCE, so the backend holds no state and never sees a token. The professional owns the archive outright: nothing to migrate off, nothing lost when a contract ends or a company folds, and a full history of every record. It also keeps the record structured from the start, so the same surveillance can export to GeoJSON for Open History Map and to a GeoContext repository, and become a published map the day an embargo lifts.

We report v1 honestly: field-first, single-archaeologist; Vincoli in Rete, ISPRA/CARG and those exports still to come; no export yet to the GNA template, the MiC standard for preventive-archaeology data. And we raise the question the panel exists for — what "open source in production" costs when the client is a public authority whose accepted deliverable is a closed document format, and what has to be conceded to stay usable in that setting.